

R&S®NRP-Zxx Power Sensors Getting Started



1176882202
Version 07

ROHDE & SCHWARZ
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This manual gives an introduction to the following R&S®NRP-Zxx power sensors:

- R&S NRP-Z2x1
- R&S NRP-Z8x
- R&S NRP-Zx1
- R&S NRP-Z9x
- R&S NRP-Z5x
- R&S NRP-Zx8

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Muehldorfstr. 15, 81671 Muenchen, Germany

Phone: +49 89 41 29 - 0

Email: info@rohde-schwarz.com

Internet: www.rohde-schwarz.com

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Throughout this document, R&S® is indicated as R&S.

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1 Safety and regulatory information

The product documentation helps you use the product safely and efficiently. Follow the instructions provided here and in the following chapters.

Intended use

The power sensors are intended for accurate and uncomplicated power measurements in production, R&D and calibration labs as well as for installation and maintenance tasks. The supported base units are listed in the data sheet. Observe the operating conditions and performance limits stated in the data sheet.

Target audience

The target audience is developers and technicians. The required skills and experience in power measurements depend on the used operating concept.

The power sensors are suitable for numerous power measurement applications. Their application depends on the power sensor type and the frequency range of the power sensor. For details, see the user manual or the data sheet.

Where do I find safety information?

Safety information is part of the product documentation. It warns you of potential dangers and gives instructions on how to prevent personal injury or damage caused by dangerous situations. Safety information is provided as follows:

- In [Section 1.1, "Safety instructions"](#), on page 5. The same information is provided in many languages as printed "Safety Instructions". The printed "Safety Instructions" are delivered with the product.
- Throughout the documentation, safety instructions are provided when you need to take care during setup or operation.

1.1 Safety instructions

Products from the Rohde & Schwarz group of companies are manufactured according to the highest technical standards. To use the products safely, follow the instructions provided here and in the product documentation. Keep the product documentation nearby and offer it to other users.

Labels on the product

Use the product only for its intended use and within its performance limits. Intended use and limits are described in the product documentation such as the data sheet, manuals and the printed "Safety Instructions". If you are unsure about the appropriate use, contact Rohde & Schwarz customer service.

Using the product requires specialists or specially trained personnel. These users also need sound knowledge of at least one of the languages in which the user interfaces and the product documentation are available.

Reconfigure or adjust the product only as described in the product documentation or the data sheet. Any other modifications can affect safety and are not permitted.

Never open the casing of the product. Only service personnel authorized by Rohde & Schwarz are allowed to repair the product. If any part of the product is damaged or broken, stop using the product. Contact Rohde & Schwarz customer service at <https://www.rohde-schwarz.com/support>.

Operating the product

Only use the product indoors. The product casing is not waterproof.

Observe the ambient conditions such as altitude, operating temperature and climatic loads; see the data sheet.

Meaning of safety labels

Safety labels on the product warn against potential hazards.



Potential hazard

Read the product documentation to avoid personal injury or product damage.

1.2 Labels on the product

Labels on the product inform about:

- Personal safety
See "Meaning of safety labels" on page 6.
- Environment safety
See Table 1-1.
- Identification of the product

Warning messages in the documentation

The name plate on the casing contains the serial number that uniquely identifies the power sensor.

Table 1-1: Labels regarding environment safety

	Labeling in line with EN 50419 for disposal of electrical and electronic equipment after the product has come to the end of its service life. For more information, see the user manual.
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1.3 Warning messages in the documentation

A warning message points out a risk or danger that you need to be aware of. The signal word indicates the severity of the safety hazard and how likely it will occur if you do not follow the safety precautions.

NOTICE

Potential risks of damage. Could result in damage to the supported product or to other property.

2 Documentation overview

This section provides an overview of the R&S NRP-Z power sensor user documentation. Unless specified otherwise, you find the documents on the R&S NRP-Z power sensor product page at:

www.rohde-schwarz.com/product/nrpz

2.1 Getting started manual

Introduces the R&S NRP-Z power sensor and describes how to set up and start working with the product. Includes basic operations and general information, e.g. safety instructions, etc. A printed version is delivered with the power sensor.

2.2 User manuals

Contains the description of all instrument modes and functions. It also provides an introduction to remote control, a complete description of the remote control commands with programming examples, and information on maintenance and interfaces. Includes the contents of the getting started manual.

2.3 Tutorials

Offer guided examples and demonstrations on operating the R&S NRP-Z power sensor. They are provided on the product page of the internet.

2.4 Instrument security procedures

Deals with security issues when working with the R&S NRP-Z power sensor in secure areas. It is available for download on the Internet.

Application notes, application cards, white papers, etc.

2.5 Data sheets and brochures

The data sheet contains the technical specifications of the R&S NRP-Z power sensor. It also lists the firmware applications and their order numbers, and optional accessories.

The brochure provides an overview of the instrument and deals with the specific characteristics.

2.6 Release notes and open-source acknowledgment (OSA)

The release notes list new features, improvements and known issues of the current firmware version, and describe the firmware installation.

The open-source acknowledgment document provides verbatim license texts of the used open source software.

2.7 Application notes, application cards, white papers, etc.

These documents deal with special applications or background information on particular topics.

3 Key features

Power measurements performed with the R&S NRP-Z power sensors can be evaluated in various ways. This manual gives a short introduction to the solutions offered by Rohde & Schwarz for viewing your power measurement results. Also, you find a description of how to set up and measure power with your power sensor.

You can view the results of your measurement on the following devices:

- PC/laptop
- Android 4.x device
- R&S NRP2 base unit
- Supported Rohde & Schwarz instrument

Using the R&S NRP-Z power sensors with a PC/laptop

Install the R&S NRP Toolkit on the PC/laptop. The R&S NRP Toolkit is a free software kit that provides drivers and tools. For details, see [Section 6.1, "R&S NRP Toolkit"](#), on page 26:

For optional use, different software solutions are offered:

- R&S Power Viewer
Free software that supports all measurement modes and can perform mathematical calculations during measurements. It can be used with one power sensor at a time.
For details, see [Section 6.3, "R&S Power Viewer"](#), on page 33.
- R&S NRPV virtual power meter
Cost-effective measurement solution that can be used as a replacement for the R&S NRP2 base unit. It supports all measurement modes and can perform different mathematical calculations during measurements. It can be used with up to four power sensors simultaneously.
For details, see [Section 6.2, "R&S NRPV"](#), on page 31.

Using the R&S NRP-Z power sensors with an Android device

Also, you can connect the power sensors to an Android device and view the results with the free of charge R&S Power Viewer Mobile.

For details, see [Section 6.4, "R&S Power Viewer Mobile"](#), on page 34.

Using the R&S NRP-Z power sensors with an R&S NRP2 or a compatible Rohde & Schwarz instrument

A short overview of the usage of the power sensors with the R&S NRP2 base unit or a compatible Rohde & Schwarz instrument is given in this manual:

- [Section 6.5, "R&S NRP2"](#), on page 34
- [Section 6.6, "Compatible instruments"](#), on page 36

For more information on this usage, refer to the operating manual of the corresponding Rohde & Schwarz instrument.

Overview of the R&S NRP-Z power sensor types

The R&S NRP-Z power sensors are highly accurate standalone measuring instruments that offer all the functionality of conventional power meters, and more, within the small housing of a power sensor. With a high dynamic range, the different types of R&S NRP-Z power sensors are suitable for nearly every measurement task:

- **Three-path power sensors**
With patented three-path technology. Suitable for many applications, thanks to their wide dynamic and frequency range. The sensors feature high measurement accuracy independent of the modulation type and speed and offer a wide range of functions.
- **Two-path power sensors**
With slightly modified dynamic range and performance. A cost-effective solution ideal for production applications.
- **Wideband power sensors**
With high video bandwidth that permits accurate time-domain analysis of the envelope power. Analysis is fast and easy thanks to also integrated measurement functions such as automatic pulse analysis.
- **Thermal power sensors**
With outstanding measurement accuracy. They are the first choice for power measurements in the frequency range from DC to 110 GHz as performed in calibration labs, where measurement accuracy is the key requirement.
- **Average power sensors**
Specially developed for EMC applications and cover the required frequency range from 9 kHz to 6 GHz.
- **Level control sensors**

With an integrated splitter that can split the signal into two parts of equal power. This feature allows you to feed power to a device under test (DUT) and monitor the power at the same time.

- Power sensor modules
Developed for level calibration using the R&S FSMR measuring receiver.

You can find an overview of the available sensors and filter according to your requirements at:

www.rohde-schwarz.com/product/nrpz

4 Preparing for use

Here, you can find basic information about setting up the product for the first time.

• Unpacking and checking.....	13
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• Powering the R&S NRP-Z power sensor.....	19

4.1 Unpacking and checking

1. Unpack the product carefully.
2. Retain the original packing material. Use it when transporting or shipping the product later.
3. Using the delivery notes, check the equipment for completeness.
4. Check the equipment for damage.

If the delivery is incomplete or equipment is damaged, contact Rohde & Schwarz.

4.2 Choosing the operating site

Specific operating conditions ensure proper operation and avoid damage to the product and connected devices. For information on environmental conditions such as ambient temperature and humidity, see the data sheet.

Electromagnetic compatibility classes

The electromagnetic compatibility (EMC) class indicates where you can operate the product. The EMC class of the product is given in the data sheet under "General data".

- Class B equipment is suitable for use in:
 - Residential environments
 - Environments that are directly connected to a low-voltage supply network that supplies residential buildings
- Class A equipment is intended for use in industrial environments. It can cause radio disturbances in residential environments due to possible conducted and radiated disturbances. It is therefore not suitable for class B environments. If class A equipment causes radio disturbances, take appropriate measures to eliminate them.

4.3 Considerations for test setup

EMI impact on measurement results

Electromagnetic interference (EMI) can affect the measurement results.

To suppress electromagnetic radiation during operation:

- Use high-quality shielded cables, for example, double-shielded RF and interface cables.
- Always terminate open cable ends.
- Ensure that connected external devices comply with EMC regulations.

Signal input and output levels

Information on signal levels is provided in the data sheet. Keep the signal levels within the specified ranges to avoid damage to the product and connected devices.

4.4 Connectors of the R&S NRP-Z power sensor

An R&S NRP-Z power sensor has two connectors, as shown in [Figure 4-1](#).



Figure 4-1: Connectors of an R&S NRP-Z power sensor

1 = Host connector

2 = RF connector

Host connector

See (1) in [Figure 4-1](#).

Lemo type host connector. You can plug it:

- Directly to a compatible Rohde & Schwarz instrument
- To a R&S NRP2 power meter base unit
- Using an R&S NRP-Z3 or R&S NRP-Z4 USB adapter to a controlling device (PC or other)

Connecting the R&S NRP-Z power sensor

- Using an R&S NRP-Z5 USB hub to a controlling device (PC or other)

RF connector

See (2) in [Figure 4-1](#).

Used to connect the power sensor to a device under test (DUT) or a signal generator.

Caused by their different frequency ranges, the R&S NRP-Z power sensors have different RF connectors. Also, for maximal measurement accuracy, the RF connector must be tightened using a torque wrench with a specific nominal torque.

Refer to the operating manual of your R&S NRP-Z power sensor for information on your connector and the allowed nominal torque.

4.5 Connecting the R&S NRP-Z power sensor

 When connecting the R&S NRP-Z27/-Z28/-Z37/-Z98 sensors, there are additional requirements to fulfill. Refer to the operating manual of the corresponding sensor for a description on how to connect it.

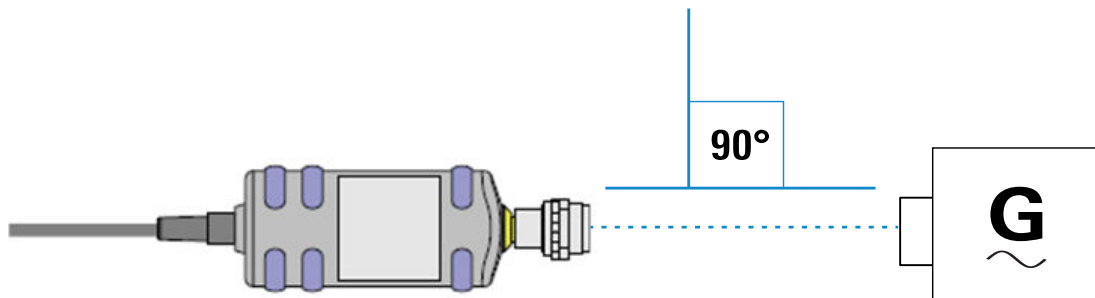
4.5.1 Connecting to a DUT

For connecting the power sensor to a DUT, use the RF connector. See "[RF connector](#)" on page 16.

To connect to a DUT

1. Ensure that the RF connector of your DUT is compatible with the RF connector of the power sensor.
2. Inspect both RF connectors carefully. Look for metal particles, contaminants and defects.
If either RF connector is damaged, do not proceed, because the risk of damaging the mating connector is too high.
3. Insert the RF connector straight into the RF output of your DUT. Take care not to tilt it.

Connecting the R&S NRP-Z power sensor



4. **NOTICE!** Risk of damaging the center pin of the RF connector. Only rotate the hex nut of the RF connector. Never rotate the power sensor itself.
Tighten the RF connector manually.
5. Tighten the RF connector using a torque wrench with the nominal torque recommended in the operating manual of your R&S NRP-Z power sensor to ensure maximum measurement accuracy.

4.5.2 Connecting to a host connector

You can use an R&S NRP-Z3 or R&S NRP-Z4 adapter.

To connect to an adapter

1. Rotate both cables, so that the red markings on the sleeves of both of them are at the same position.
2. Insert the host connector of your sensor into the connector of your USB adapter.

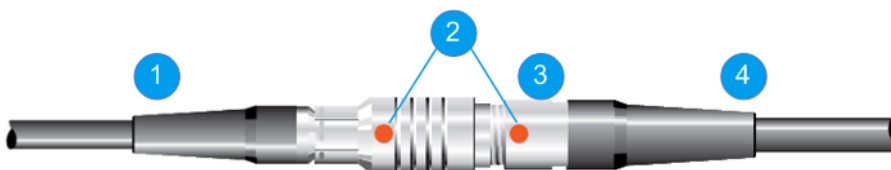


Figure 4-2: Connecting the host connector

- 1 = R&S NRP-Z4 in this example
- 2 = Red marking
- 3 = Host connector
- 4 = R&S NRP-Z power sensor

Disconnecting the R&S NRP-Z power sensor

To connect to an R&S NRP-Z5 sensor hub

- ▶ Insert the host connector, red marking upwards, into the SENSOR connector of your sensor hub.

To connect to a Rohde & Schwarz instrument

- ▶ Insert the host connector, red marking upwards, into the SENSOR connector of your instrument.

4.6 Disconnecting the R&S NRP-Z power sensor

 Additional requirements must be fulfilled when disconnecting the R&S NRP-Z27/-Z28/-Z37/-Z98 sensors. Refer to the operating manual of the corresponding sensor for a description on how to disconnect it.

4.6.1 Disconnecting from a DUT

To disconnect from a DUT

1. **NOTICE!** Risk of damaging the center pin of the RF connector. Only rotate the hex nut of the RF connector. Never rotate the power sensor itself.
Carefully loosen the union nut at the front of the RF connector of the power sensor.
2. Remove the power sensor.

4.6.2 Disconnecting from a host connector

To disconnect from an R&S NRP-Z3 or R&S NRP-Z4 adapter

- ▶ Pull at the host connector (3 in [Figure 4-2](#)) until it disconnects from the connector of the USB adapter.

Powering the R&S NRP-Z power sensor

To disconnect from a Rohde & Schwarz instrument or an R&S NRP-Z5 sensor hub

- At the same time, press down the ring of the built-in plug (1) and pull off the cable sleeve (2).

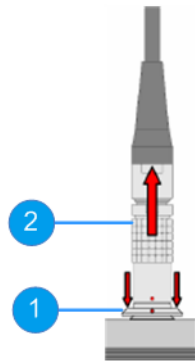


Figure 4-3: Disconnecting from a SENSOR connector

1 = Ring of the built-in plug
2 = Cable sleeve

4.7 Powering the R&S NRP-Z power sensor

The R&S NRP-Z power sensors can be powered in two ways:

- Self-powered from a separate power supply using the R&S NRP-Z3 active USB adapter or the R&S NRP-Z5 sensor hub.
- Bus-powered from the PC or a USB hub with its own power supply (self-powered hub) using the R&S NRP-Z3 active USB adapter or the R&S NRP-Z4 passive USB adapter.

Requirements for the usage in the bus-powered mode:

- Laptop/PC with a self-powered USB hub
- Total available current per USB port of 500 mA

5 Connecting to a PC

There are different possibilities for connecting a power sensor to a PC, which can differ depending on the type of power sensor and available accessories.

The following chapter gives an overview of the possible setups and what equipment is needed for each of them.

For details on powering the R&S NRP-Z power sensor, see [Section 4.7, "Powering the R&S NRP-Z power sensor"](#), on page 19.

5.1 Using a passive USB adapter

The R&S NRP-Z power sensors can be used with the R&S NRP-Z4 for power measurements where no external triggering is required.

Required equipment

- R&S NRP-Z power sensor
- R&S NRP-Z4 passive USB adapter
- PC with installed R&S NRP Toolkit, see [Section 6.1, "R&S NRP Toolkit"](#), on page 26

Setup

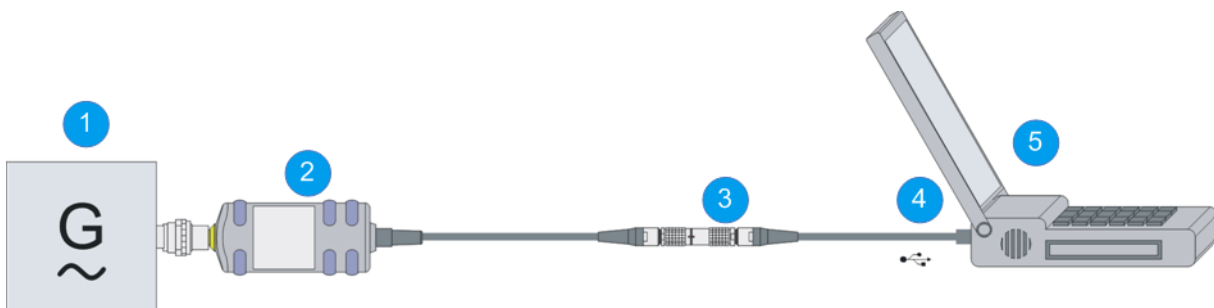


Figure 5-1: Setup with a passive USB adapter

- 1 = Signal source
- 2 = R&S NRP-Z power sensor
- 3 = R&S NRP-Z4
- 4 = USB connector
- 5 = PC with a USB interface

 Incorrectly connecting/disconnecting the R&S NRP-Z power sensors may damage the power sensors or lead to erroneous results.

Ensure that you connect/disconnect your power sensors as described in [Section 4.5, "Connecting the R&S NRP-Z power sensor"](#), on page 16 and [Section 4.6, "Disconnecting the R&S NRP-Z power sensor"](#), on page 18.

1. Connect the cables as shown in [Figure 5-1](#):
 - a) Connect the R&S NRP-Z4 passive USB adapter to the power sensor.
 - b) Connect the power sensor passive USB adapter to the computer.
 - c) Connect the power sensor to the signal source.
2. On the PC, start a software application to view the measurement results. See [Section 6, "Operating concepts"](#), on page 26.

5.2 Using an active USB adapter

The R&S NRP-Z power sensors can be used with the R&S NRP-Z3 for power measurements where an external power supply and/or external triggering is required.

Required equipment

- R&S NRP-Z power sensor
- R&S NRP-Z3
- PC with installed R&S NRP Toolkit, see [Section 6.1, "R&S NRP Toolkit"](#), on page 26

Setup

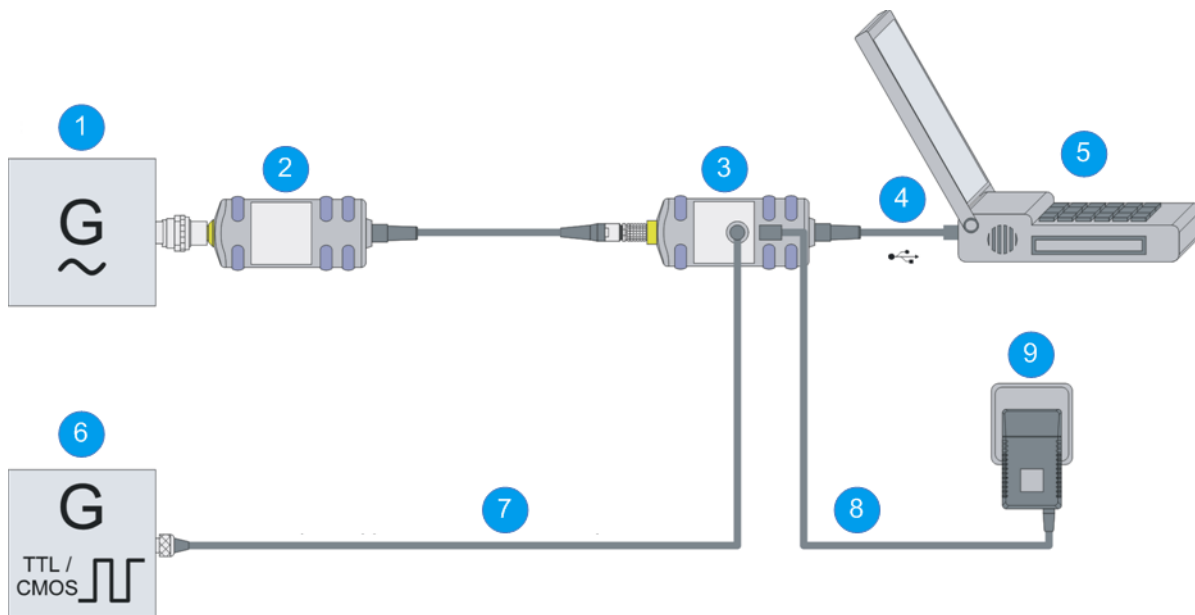


Figure 5-2: Setup with an active USB adapter

- 1 = Signal source
- 2 = R&S NRP-Z power sensor
- 3 = R&S NRP-Z3
- 4 = USB connector
- 5 = PC with a USB interface
- 6 = Trigger source (optional)
- 7 = BNC cable (optional, not supplied)
- 8 = Plug-in power supply (optional, supplied)
- 9 = AC supply connector (optional)

i Incorrectly connecting/disconnecting the R&S NRP-Z power sensors may damage the power sensors or lead to erroneous results.

Ensure that you connect/disconnect your power sensors as described in [Section 4.5, "Connecting the R&S NRP-Z power sensor"](#), on page 16 and [Section 4.6, "Disconnecting the R&S NRP-Z power sensor"](#), on page 18.

1. Connect the cables as shown in [Figure 5-2](#):
 - a) Connect the R&S NRP-Z3 active USB adapter to the power sensor.
 - b) Connect the power sensor passive USB adapter to the computer.
 - c) Connect the power sensor to the signal source.
 - d) Connect the R&S NRP-Z3 with a BNC cable to the trigger source (optional).

Using an R&S NRP-Z5 sensor hub

- e) Connect the delivered plug-in power supply to the R&S NRP-Z3 and to an AC supply connector (optional).
2. On the PC, start a software application to view the measurement results. See [Section 6, "Operating concepts"](#), on page 26.

5.3 Using an R&S NRP-Z5 sensor hub

The R&S NRP-Z5 sensor hub (high-speed USB 2.0) can host up to four R&S NRP-Z power sensors and provides simultaneous internal and external triggering of all connected sensors.

Required equipment

- R&S NRP-Z power sensor
- R&S NRP-Z5 sensor hub that provides:
 - Power supply
 - Through-wired trigger bus
 - Trigger input and trigger output via BNC sockets
- PC with installed R&S NRP Toolkit, see [Section 6.1, "R&S NRP Toolkit"](#), on page 26

Using an R&S NRP-Z5 sensor hub

Setup

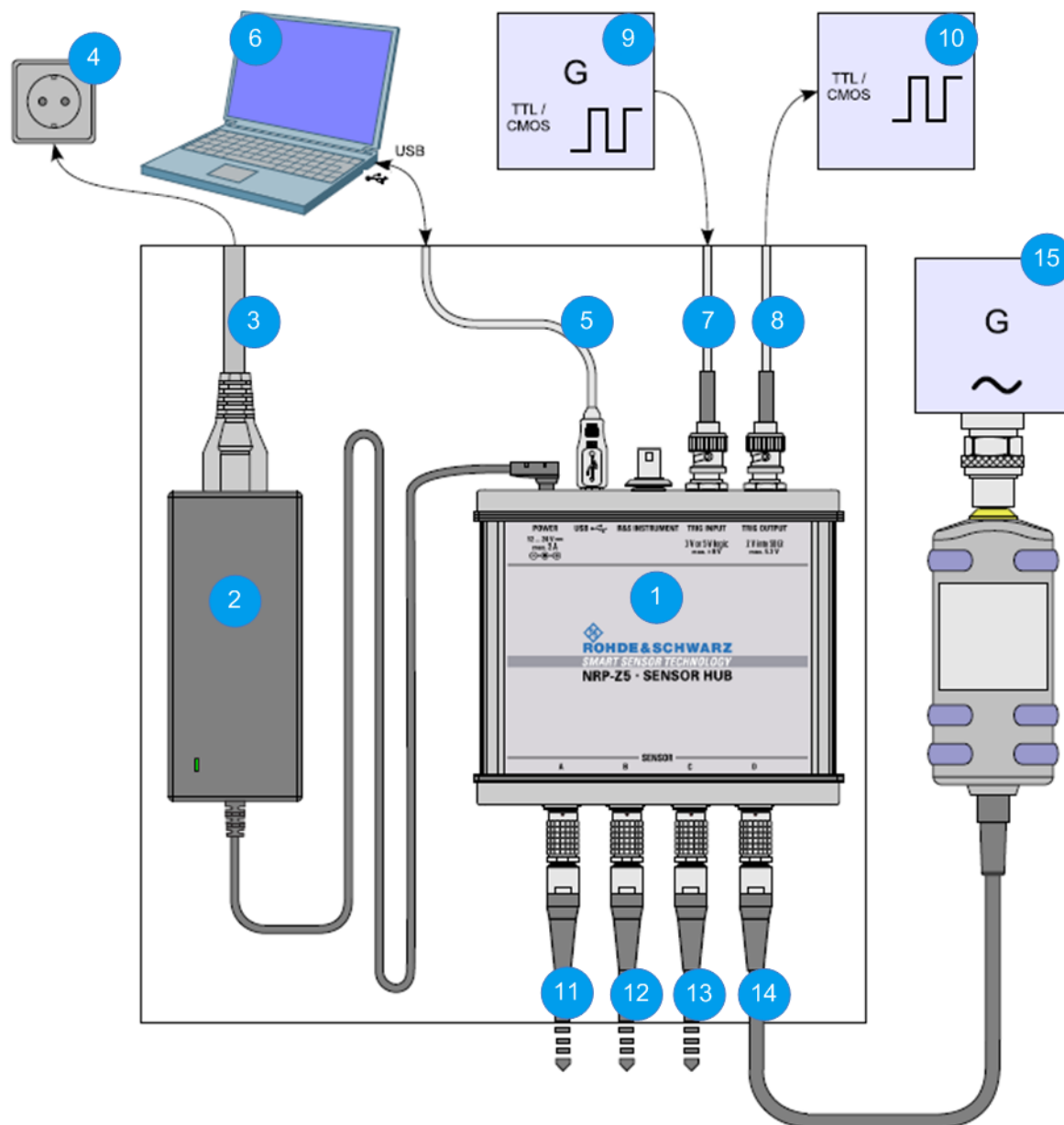


Figure 5-3: Setup with an R&S NRP-Z5 sensor hub

- 1 = R&S NRP-Z5 sensor hub
- 2 = External power supply unit (supplied)
- 3 = Power cable (supplied)
- 4 = AC power supply
- 5 = USB cable (supplied)
- 6 = PC with USB host interface
- 7, 8 = BNC cable (optional, not supplied)
- 9 = Trigger source (optional)

Using an R&S NRP-Z5 sensor hub

- 10 = Triggered device (optional)
- 11-14 = R&S NRP-Z power sensor
- 15 = Signal source

 Incorrectly connecting/disconnecting the R&S NRP-Z power sensors may damage the power sensors or lead to erroneous results.

Ensure that you connect/disconnect your power sensors as described in [Section 4.5, "Connecting the R&S NRP-Z power sensor"](#), on page 16 and [Section 4.6, "Disconnecting the R&S NRP-Z power sensor"](#), on page 18.

1. Connect the cables as shown in [Figure 5-3](#):
 - a) Connect the power sensors to the R&S NRP-Z5 sensor hub. You can connect up to four sensors.
 - b) Connect the R&S NRP-Z5 to the computer.
 - c) Connect the power sensors to the signal source.
 - d) Connect the delivered external power supply unit to the R&S NRP-Z5 and to an AC supply connector.
 - e) Connect the R&S NRP-Z5 with a BNC cable to the trigger source (optional).
 - f) Connect the R&S NRP-Z5 with a BNC cable to the trigger device (optional).
2. On the PC, start a software application to view the measurement results. See [Section 6, "Operating concepts"](#), on page 26.

6 Operating concepts

For operating the power sensor, you can choose from various possibilities.

Alternative to the methods of access described in this chapter, you can use remote control. For details, see:

- [Section 7, "Remote control"](#), on page 39
- Operating manual of the R&S NRP-Z power sensor
- [R&S NRP Toolkit](#)..... 26
- [R&S NRPV](#)..... 31
- [R&S Power Viewer](#)..... 33
- [R&S Power Viewer Mobile](#)..... 34
- [R&S NRP2](#)..... 34
- [Compatible instruments](#)..... 36

6.1 R&S NRP Toolkit

 Before you start using the power sensors with a software solution, you have to install the R&S NRP Toolkit.

The R&S NRP Toolkit is the basic software package that supplies low-level drivers and tools for all power sensors. It is provided on your documentation CD-ROM and on the Rohde & Schwarz website. The content of the toolkit depends on your operating system.

 To ensure that your power sensor is recognized properly by your system, first install the R&S NRP Toolkit on your laptop/PC and then connect your power sensor for the first time.

6.1.1 System requirements

Hardware requirements:

- Desktop PC or laptop, or an Intel-based Apple Mac
- USB interface

- R&S NRP-Z3 or R&S NRP-Z4 USB adapter or R&S NRP-Z5 sensor hub

Supported operating systems:

- Microsoft Windows Vista 32/64-bit
- Microsoft Windows 7 32/64-bit
- Microsoft Windows 8 32/64-bit
- Microsoft Windows 10 32/64-bit
- Microsoft Windows XP 32-bit is available on request only.

R&S NRP Toolkit versions for Linux distributions and MacOSX are also available on request. To obtain an R&S NRP Toolkit for an operating system other than Microsoft Windows, contact the Rohde & Schwarz customer support:

customersupport@rohde-schwarz.com

6.1.2 Windows operating system

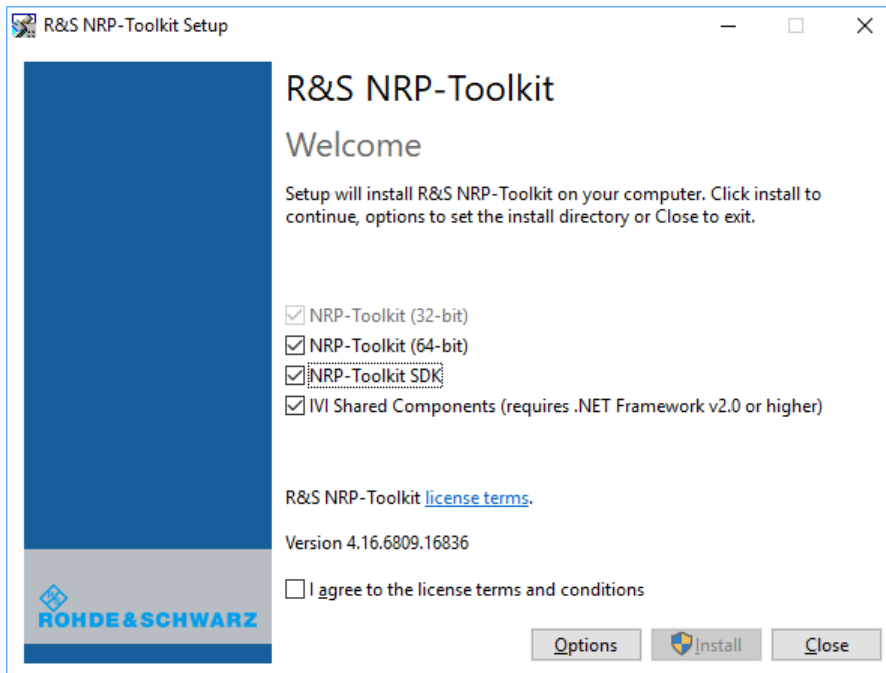
The R&S NRP Toolkit installer version for Windows-based systems contains the following components. Obligatory components are indicated.

- USB driver (obligatory)
- High level dynamic link library (NrpControl2) (obligatory)
- R&S NRP Toolkit SDK
- S-Parameter tool
Program modules for loading an S-parameter table into the power sensor
- R&S NRP-Z uncertainty calculator and its PDF manual
- Vxl plug&play with programming examples
For installation, enable "NRP-Toolkit-SDK" during the installation of the R&S NRP Toolkit.
- Terminal (NrpTerm): low-level communication program for sending commands to the power sensor
- Firmware update (PureFW)
- Nrp Version Collector
Tool for displaying version information of all installed, power measurement-relevant software packages.

To install the R&S NRP Toolkit

1. Start the R&S NRP Toolkit installer.

2. In the "Choose Components" dialog, select the components you want to install and accept the license terms to continue with the installation



3. Click "Next" and complete the installation process.

6.1.3 Linux-based systems

The R&S NRP Toolkit for a certain Linux distribution comes as a series of Debian (*.deb) or RPM (*.rpm) files. These files become installed through the standard installation process of the individual Linux distribution. See the `ReadMe` file that accompanies the R&S NRP Toolkit for further instructions.

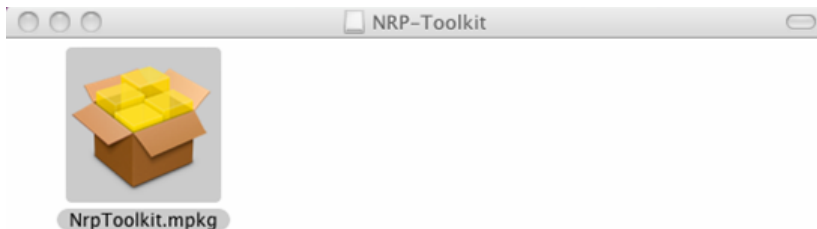
6.1.4 Mac OS X

The R&S NRP Toolkit installer version for Mac OS X contains the following components:

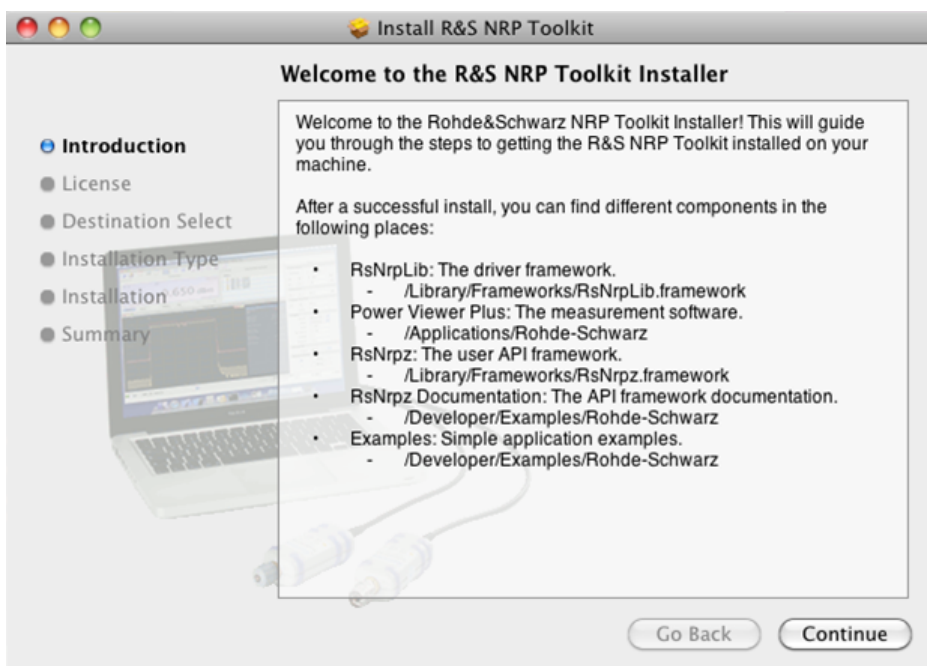
- Low-level driver: `RsNrpLib.framework`
- VXI PnP driver: `RsNrpz.framework`
- HTML help files for the VXI PnP driver
- Power Viewer Plus and its PDF manual
- Example programs for use with VXI PnP driver

To install the R&S NRP Toolkit

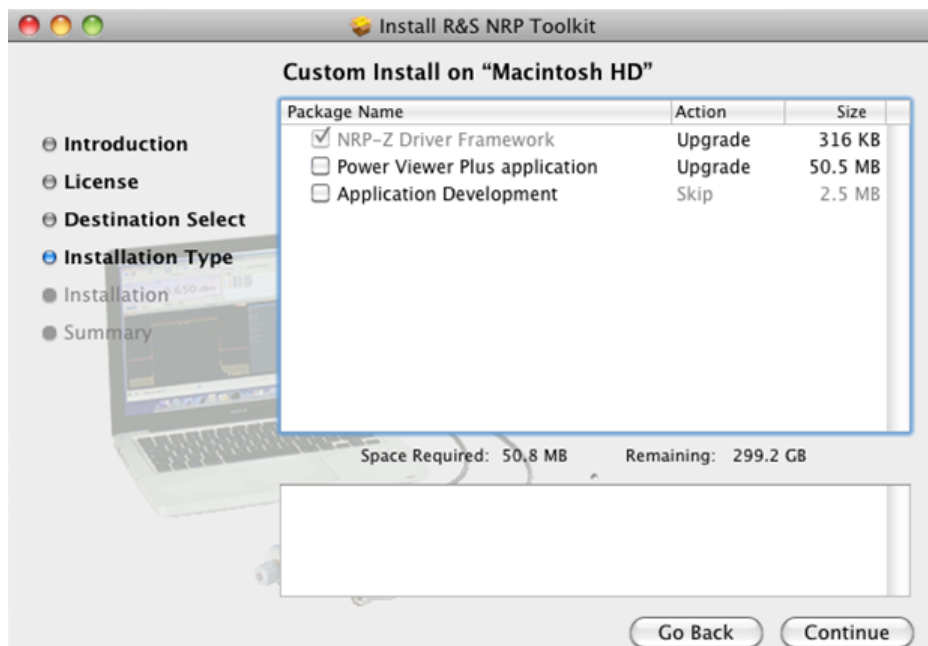
1. In the Mac OS X Finder, double-click the provided .dmg disk image to mount it.
2. Double-click the `NrpToolkit.mpkg` installer to start the installation.



The R&S NRP Toolkit installer opens. The welcome message gives an overview of the packages that are part of the installer and indicates their default installation location.



3. Accept the license terms to continue with the installation.
4. Select the components you want to install and click "Continue" to complete the installation process.



- a) R&S NRP-Z Driver Framework (obligatory).
 - b) Enable the "Power Viewer Plus application" (optional).
 - c) Enable "Application Development" (optional). You can use these examples as starting points for your own implementations.
5. Click "Continue" and complete the installation process.

After a successful installation, you can start the applications provided with the R&S NRP Toolkit from the Rohde-Schwarz folder that was created in the Mac OS X application directory.



6.1.5 Performing a firmware update

Use the firmware update program (PureFW) to load new firmware for the power sensors. It is part of the R&S NRP Toolkit that is supplied on a CD-ROM together with the power sensors. For further details, refer to the operating manual.

6.2 R&S NRPV

The R&S NRPV enables you to measure power in all available measurement modes. Also, you can use up to four power sensors simultaneously.

The R&S NRPV software can be installed on any Windows PC. It is provided on your documentation CD-ROM and on the Rohde & Schwarz website as a separate standalone installation package. To make full use of its functionalities, you must activate your R&S NRP-Z power sensor with a license key.

Required equipment

- R&S NRP-Z power sensor
- R&S NRP-Z3, R&S NRP-Z4 or R&S NRP-Z5
- Windows PC with installed:
 - R&S NRP Toolkit
 - R&S NRPV (refer to the operating manual of the R&S NRPV for a description of the installation process)
- License key (R&S NRPZ-K1 option) for activating the power measurements for your R&S NRP-Z power sensor

Setup

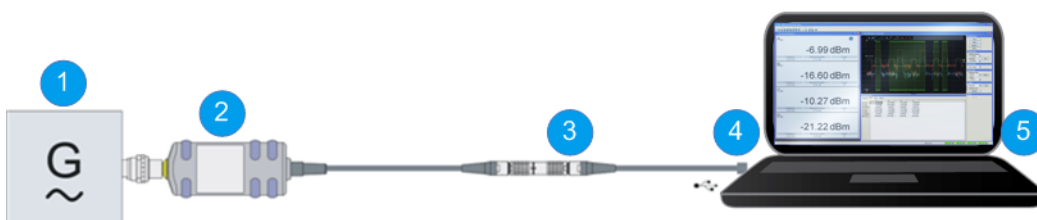


Figure 6-1: Setup with an R&S NRP-Z4 and an R&S NRPV

- 1 = Signal source
- 2 = R&S NRP-Z power sensor
- 3 = R&S NRP-Z4
- 4 = USB connector
- 5 = PC with installed R&S NRPV

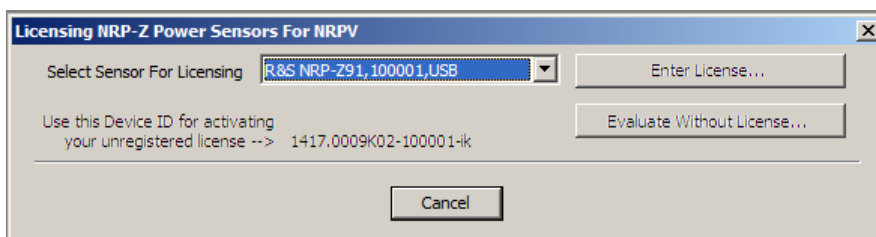
 Incorrectly connecting/disconnecting the R&S NRP-Z power sensors may damage the power sensors or lead to erroneous results.

Ensure that you connect/disconnect your power sensors as described in [Section 4.5, "Connecting the R&S NRP-Z power sensor"](#), on page 16 and [Section 4.6, "Disconnecting the R&S NRP-Z power sensor"](#), on page 18.

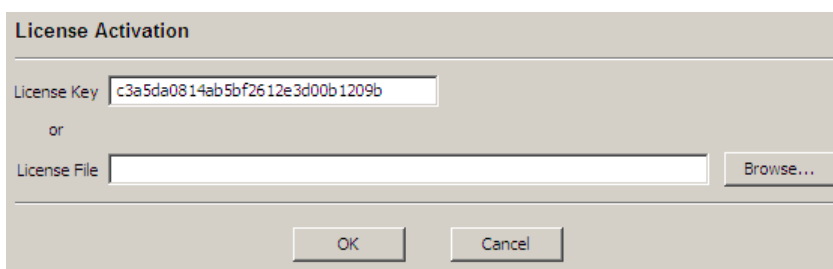
Starting a measurement

1. Connect the power sensor to the PC as shown in [Figure 6-1](#).
2. On the PC, start the R&S NRPV software.
3. Connect the power sensor to the PC. For a detailed description, see [Section 5, "Connecting to a PC"](#), on page 20.
4. If you use the power sensor for the first time with the R&S NRPV software, activate the sensor using a license key:
 - a) In the "File" menu, select "File" > "Licensing".

The "Licensing NRP-Z Power Sensors for NRPV" dialog opens.



- b) Select your power sensor.
 - c) Select "Enter License".
- The "License Activation" dialog opens.



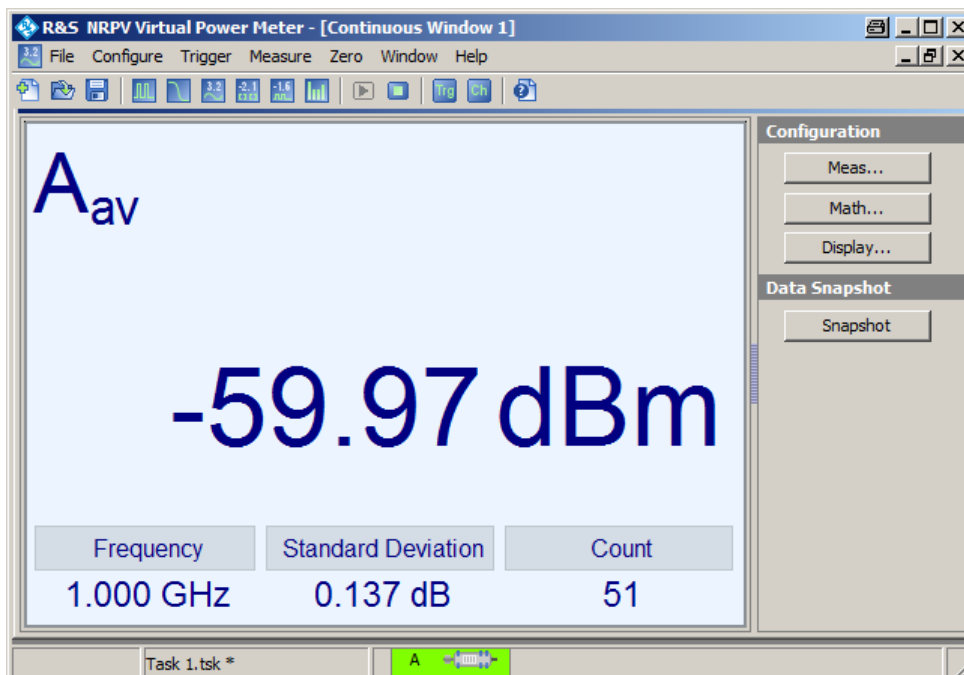
- d) Enter the license key either manually or with the key code file.

- e) Confirm with "OK" to return to the "Licensing NRP-Z Power Sensors for NRPV" dialog. The dialog indicates all currently connected sensors.

Note: If you want to evaluate the R&S NRPV before buying a license for your power sensor, you can activate a sensor for temporary use. To do so, select "Evaluate Without License" in the "Licensing NRP-Z Power Sensors for NRPV" dialog. For a detailed description, refer to the operating manual of the R&S NRPV.

5. Switch on the test signal of the signal source.
6. To start a continuous measurement, select "Measure" > "Continuous".

The "Continuous" measurement window appears. It shows the measurement results numerically, and the control panel for accessing further dialogs with parameters for measurement, evaluation and display.



For a detailed description of how to measure in this setup, refer to the operating manual of the R&S NRPV.

6.3 R&S Power Viewer

The R&S Power Viewer is software that simplifies many measurement tasks. It is provided on your documentation CD-ROM and on the Rohde & Schwarz website as a separate standalone installation package.

For details on installing and operating the R&S Power Viewer, refer to its operating manual. The manual is provided on the documentation CD-ROM and is installed automatically during the installation of the R&S Power Viewer.

 If you want to use an android device like a tablet or a smartphone, use the R&S Power Viewer Mobile. For details, see [Section 6.4, "R&S Power Viewer Mobile"](#), on page 34.

6.4 R&S Power Viewer Mobile

The R&S Power Viewer Mobile extends the functionality of the R&S Power Viewer to Android-based devices, such as a smartphone and tablets.

 For connecting the power sensor to USB-C type mobile phones (Android), use an R&S NRP-ZKC cable. It enables the R&S Power Viewer Mobile to take power measurements via the USB-C connection.

You can download the R&S Power Viewer Mobile free of charge from the Google Play Store.

The 1MA215 "Using R&S®NRP Series Power Sensors with Android™ Handheld Devices" application note gives a detailed description on installation and features of the R&S Power Viewer Mobile. The application note is provided on the documentation CD-ROM and at:

www.rohde-schwarz.com/application/nrpz

6.5 R&S NRP2

With the R&S NRP-Z power sensors and an R&S NRP2, you can measure power with up to four power sensors simultaneously, depending on the base unit configuration. All sensor-dependent measurement functions can be used and the results can be displayed in parallel.

Required equipment

- R&S NRP-Z power sensor

- R&S NRP2 base unit

Setup



Figure 6-2: Setup with an R&S NRP2 base unit

- 1 = Signal source
- 2 = R&S NRP-Z power sensor
- 3 = SENSOR connector of the R&S NRP2
- 4 = R&S NRP2 base unit

i Incorrectly connecting/disconnecting the R&S NRP-Z power sensors may damage the power sensors or lead to erroneous results.

Ensure that you connect/disconnect your power sensors as described in [Section 4.5, "Connecting the R&S NRP-Z power sensor"](#), on page 16 and [Section 4.6, "Disconnecting the R&S NRP-Z power sensor"](#), on page 18.

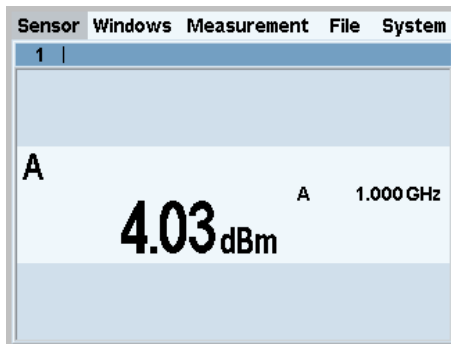
Starting a measurement

1. Connect the power sensor to the [SENSOR] connector of the R&S NRP2 as shown in [Figure 6-2](#).
2. Preset the R&S NRP2.
3. Execute zeroing.

Note: Turn off all test signals before zeroing. An active test signal during zeroing causes an error.

4. Connect the power sensor to the signal source.

The result window indicates the result (in dBm) obtained with sensor A.



5. If necessary, perform further settings.

For a detailed description of how to measure in this setup, refer to the operating manual of your R&S NRP2.

6.6 Compatible instruments

Many Rohde & Schwarz instruments allow power measurements using the R&S NRP-Z power sensors. For example, the power sensors can be used with a signal generator to achieve stable and accurate RF power, which can be further supplied to your DUT.

You can connect the power sensors to the following instrument connectors:

- Directly to the SENSOR connector
- To the USB port of the instrument, using an R&S NRP-Z3 or R&S NRP-Z4 USB adapter
- To the USB port of the instrument, using an R&S NRP-Z5 USB hub

Refer to the operating manual of your Rohde & Schwarz instrument to find out if it supports power measurements with the R&S NRP-Z power sensors.

Required equipment

- R&S NRP-Z power sensor
- Compatible Rohde & Schwarz instrument
- R&S NRP-Z3 or R&S NRP-Z4 USB adapter or an R&S NRP-Z5 sensor hub (optional)

Setup



Figure 6-3: Setup with an R&S SMF

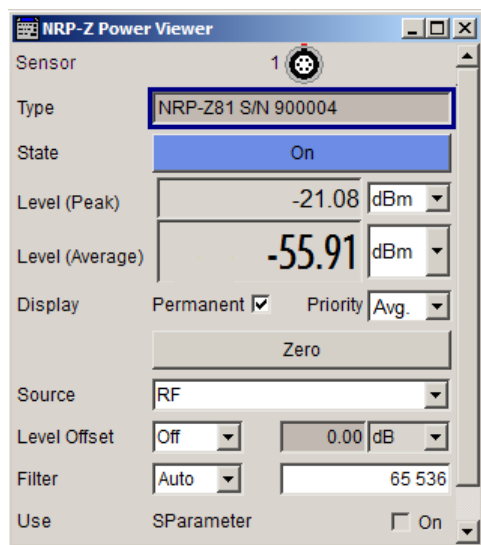
- 1 = R&S SMF
- 2 = Device under test (DUT)
- 3 = R&S NRP-Z icon
- 4 = [SENSOR] connector
- 5 = R&S NRP-Z Power Sensors
- 6 = RF output of the DUT
- 7 = [RF OUT] connector of the R&S SMF
- 8 = RF input of the DUT

i Incorrectly connecting/disconnecting the R&S NRP-Z power sensors may damage the power sensors or lead to erroneous results.

Ensure that you connect/disconnect your power sensors as described in [Section 4.5, "Connecting the R&S NRP-Z power sensor"](#), on page 16 and [Section 4.6, "Disconnecting the R&S NRP-Z power sensor"](#), on page 18.

Starting a measurement

1. Connect the cables as shown in [Figure 6-3](#):
 - a) Connect the DUT to the [RF OUT] connector of the R&S SMF.
 - b) Connect the DUT to the [RF] connector of the power sensor.
 - c) Connect the power sensor to the [SENSOR] connector of the R&S SMF.
2. On the R&S SMF, open the "NRP-Z Power Viewer" dialog.



The dialog shows the result (in dBm) and some additional parameters.

3. If necessary, perform further settings.

For a detailed description of how to measure in this setup, refer to the operating manual your compatible Rohde & Schwarz instrument.

7 Remote control

Use remote control to integrate the R&S NRP-Z power sensors into custom automatic test equipment (ATE) systems. The latest version of the available instrument drivers is part of the R&S NRP Toolkit SDK under Windows.

For a detailed description on how to perform different remote measurements, refer to various sample programs installed with the R&S NRP Toolkit SDK, or the application note "1GP69: R&S®NRP-Z Power Sensor Programming Guide".

8 Contacting customer support

Technical support – where and when you need it

For quick, expert help with any Rohde & Schwarz product, contact our customer support center. A team of highly qualified engineers provides support and works with you to find a solution to your query on any aspect of the operation, programming or applications of Rohde & Schwarz products.

Contact information

Contact our customer support center at www.rohde-schwarz.com/support, or follow this QR code:



Figure 8-1: QR code to the Rohde & Schwarz support page

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